

# INT104 期末备考笔记（针对考试版）

考试结构按你提供的信息整理：28 个选择题 = 56 分；3 道计算题 = 8 + 15 + 15 分；1 道回答题 = 6 分。

本资料根据文件夹中的每周笔记、历年题与 sample questions 整理；Python 编程/填空题不纳入复习重点。

## 重要更新：公式提供规则

根据课程提醒：

- 机器学习与模式识别相关公式不会提供，例如 decision trees、Naive Bayes、confusion matrix 等。这类公式必须背，并且要能解释每个术语的含义。
- 数学/统计相关公式会提供，例如 correlation coefficient、least squares for linear regression 等。这类公式不用死背完整形式，但必须会解释变量含义、会代入计算、会说明结论。

因此复习时不要平均用力：统计公式重点练“代入和解释”，机器学习公式重点练“默写、解释、手算”。

## 必背公式总表（不会提供）

### A. Confusion Matrix 与分类评估

四个基本量必须会解释：

TP: actual positive, predicted positive

TN: actual negative, predicted negative

FP: actual negative, predicted positive

FN: actual positive, predicted negative

必须背：

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$
$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$
$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$
$$\text{F1} = 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})$$
$$\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$$
$$\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$$

必须会解释：

- Accuracy：整体预测正确比例。
- Precision：预测为正的样本中，有多少是真的正。
- Recall：真实为正的样本中，有多少被找出来。
- F1：precision 与 recall 的调和平均。
- AUC = 0.5 表示随机分类器水平。

- Class imbalance 时 accuracy 可能误导。

## B. Naive Bayes

必须背分类规则：

Choose class  $C$  that maximizes:

$$P(C) * P(x_1 | C) * P(x_2 | C) * \dots * P(x_d | C)$$

也就是：

$$\operatorname{argmax}_C P(C | x)$$

$$\text{proportional to } \operatorname{argmax}_C P(C) * \text{product } P(x_i | C)$$

必须会解释：

- Prior,  $P(C)$ ：类别本身出现的概率。
- Likelihood / class-conditional probability,  $P(x_i | C)$ ：在某类别下某特征值出现的概率。
- Posterior,  $P(C | x)$ ：看到样本特征后属于某类别的概率。
- Naive assumption：给定类别后，各特征条件独立。

Laplace smoothing 必须背：

$$P(\text{feature value} | \text{class})$$

$$= (\text{count} + 1) / (\text{class total} + \text{number of possible feature values})$$

作用：解决 zero-probability issue。

## C. Decision Tree

Entropy 必须背：

$$\text{Entropy}(S) = - \sum p_i \log_2(p_i)$$

Information Gain 必须背：

$$\text{Gain}(S, A) = \text{Entropy}(S) - \text{weighted entropy after splitting by } A$$

Gini Impurity 必须背：

$$\text{Gini}(S) = 1 - \sum p_i^2$$

必须会解释：

- Entropy 越低，节点越纯。
- Information gain 越大，属性越适合作为划分特征。
- Gini 越小，节点越纯。
- Leaf node 中有多个类别时，通常使用 majority vote。

## D. kNN 与距离

City-block / Manhattan distance 必须背:

$$d(x, z) = \sum |x_i - z_i|$$

Euclidean distance 必须背:

$$d(x, z) = \sqrt{\sum (x_i - z_i)^2}$$

必须会解释:

- kNN 通过最近的 k 个训练样本投票分类。
- k 太小容易过拟合, k 太大容易欠拟合。
- 特征尺度会影响距离, 所以 feature scaling 很重要。

## E. k-means

分配规则必须会写:

Assign each sample to the nearest centroid.

中心更新必须背:

new centroid = mean of all samples in the cluster

目标函数了解:

minimize sum of squared distances from samples to their assigned centroids

必须会解释:

- k 是需要提前指定的 hyper-parameter。
- 初始中心会影响结果。
- k-means 对噪声和 outliers 敏感。

## F. Hierarchical Clustering

必须会解释 linkage:

Single linkage = minimum distance between two clusters

Complete linkage = maximum distance between two clusters

Average linkage = average pairwise distance between two clusters

Centroid linkage = distance between centroids

必须会做:

1. 先把每个点看成一个 cluster。
2. 每次合并距离最近的两个 clusters。

3. 记录合并距离并画 dendrogram。

## G. Silhouette Coefficient

必须背：

$$s = (b - a) / \max(a, b)$$

必须会解释：

- a: 样本到同一 cluster 内其他样本的平均距离。
- b: 样本到最近其他 cluster 的平均距离。
- s 越接近 1, 聚类越好; 接近 0 表示重叠; 小于 0 表示可能分错。

## H. SVM

必须会解释：

- Hyperplane: 分类边界。
- Margin: 分类边界到最近样本的间隔。
- Support vectors: 决定边界位置的关键样本。
- Hard margin: 不允许分类错误或越界。
- Soft margin: 允许少量 margin violation。
- C: 控制违反 margin 的惩罚。C 大更严格, C 小更宽松。
- Kernel: 把数据映射到高维以处理非线性分类。

建议会写：

Hard-margin SVM minimizes  $1/2 ||w||^2$  to maximize the margin.

## I. Cross Validation

必须会解释：

K-fold CV: split data into K folds; each fold is used once as validation/test data, and the remaining folds are used for training.

必须会算：

fold accuracy = correct predictions in the fold / number of test samples in the fold  
overall CV accuracy = average of fold accuracies

## J. Ensemble Learning

必须会解释：

- Hard voting: 每个分类器投类别, 票数最多的类别胜出。
- Soft voting: 综合各分类器的预测概率, 概率最高的类别胜出。
- Random forest: 多个 decision trees 的 ensemble。

- Bagging: 有放回抽样训练多个模型, 再聚合结果。

## K. GMM 与模型选择

必须会解释:

- GMM 用多个 Gaussian components 混合表示数据分布。
- Mixing weight: 某个 component 在整体分布中的权重。
- Mean: component 的中心。
- Covariance matrix: component 的形状、方向和尺度。
- Diagonal covariance: 特征不相关, 但不同特征可以有不同方差。
- GMM 是 soft assignment; k-means 是 hard assignment。

建议记住答题公式:

$$\begin{aligned} \text{AIC} &= 2k - 2 \ln(L) \\ \text{BIC} &= k \ln(n) - 2 \ln(L) \end{aligned}$$

其中  $k$  是参数数量,  $n$  是样本量,  $L$  是最大似然。AIC/BIC 都是越小越好, BIC 对复杂模型惩罚更强。

## 会提供但必须会用的统计公式

这些公式考试会提供, 但仍要会解释变量含义、会代入、会写结论:

Pearson correlation coefficient  
Least-squares slope and intercept for simple linear regression  
Z-score / normal distribution related formulae  
Confidence interval and hypothesis testing formulae  
Mean, variance, standard deviation  
Quartiles, IQR, 1.5\*IQR outlier rule

统计题复习重点:

1. 看懂题目变量。
2. 正确代入公式。
3. 解释结果, 例如  $r$  的方向和强度、残差正负、 $p$ -value 与  $\alpha$  的比较。
4. 画图题要会说明分布形状、偏态、outliers。

## 0. 复习优先级

### 必保模块

1. 统计基础与数据分析: 正态分布、CLT、置信区间、假设检验、Z/T、相关系数、直方图、箱线图、五数概括。
2. 线性回归: 最小二乘、斜率、截距、残差、预测。
3. kNN: City-block/Manhattan 距离、1-NN/3-NN、交叉验证、accuracy。
4. Naive Bayes: 先验、条件概率、后验、条件独立、Laplace smoothing。

5. 聚类: k-means、层次聚类、dendrogram、silhouette、elbow method。
6. 分类评估: confusion matrix、accuracy、precision、recall、F1、ROC/AUC、class imbalance。
7. SVM: margin、support vector、hard/soft margin、C、kernel、OvO/OvR。
8. 决策树与集成学习: entropy、information gain、Gini、random forest、hard/soft voting。
9. GMM 与模型选择: AIC/BIC、cross validation、Gaussian assumption、covariance matrix。
10. AI 基础与安全: training data、label、supervised/unsupervised、federated learning、data poisoning、AI 类型。

## 大题押题方向

按照 25-26 sample 和近年题目, 最值得准备:

1. 8 分计算: simple linear regression 手算。
2. 15 分计算: histogram + cumulative distribution + box plot + outlier + scaling scheme。
3. 15 分计算: kNN with fixed-fold cross validation, 尤其 City-block distance。
4. 6 分回答: GMM 模型选择标准 vs cross validation, 或模型评估/过拟合/accuracy 局限。

旧题高频但今年未必作为大题: Naive Bayes golf table、k-means one iteration、hierarchical clustering dendrogram。建议会做, 不需要投入到 Python 题那种程度。

## 1. 选择题高频知识点

### 1.1 统计基础

#### Statistic vs Parameter

- Statistic: 描述 sample 的数值, 例如样本均值。
- Parameter: 描述 population 的数值, 例如总体均值。

#### Sampling bias

- Voluntary response sampling 容易有偏, 因为愿意主动回应的人不一定代表总体。
- Random sampling、stratified sampling、cluster sampling 通常是为了降低偏差。

#### Normal distribution

- Continuous and symmetric。
- 均值、众数、中位数相同。
- 标准正态分布是  $N(0, 1)$ 。

#### Central Limit Theorem, CLT

- 大样本下, 样本均值的抽样分布近似正态。
- 常见条件: independent、identically distributed、large sample size。
- 不要求 population 本身一定是 normal。

#### Z-score

公式:

$$z = (x - \text{mean}) / \text{standard deviation}$$

Z-score 依赖 raw score、population mean、population standard deviation。

### Z-distribution vs t-distribution

- Z: 大样本, 或总体标准差已知。
- t: 小样本, 或总体标准差未知。
- t 分布比标准正态分布尾部更厚。

### Confidence interval

- 用一个区间估计总体参数。
- 不是“参数有 95% 概率在区间内”，更准确说法是：长期重复抽样时，95% 的这类区间会覆盖真实参数。

### Hypothesis testing

- p-value  $\leq$  alpha: reject  $H_0$ 。
- p-value  $>$  alpha: fail to reject  $H_0$ 。

## 1.2 概率、MLE、MAP

### Likelihood

投硬币 3 heads, 2 tails, 若模型参数是  $P(H)=p$ :

$$L(p) = p^3 (1-p)^2$$

如果题目不考虑排列组合，直接写每次观测概率的乘积。

### MLE vs MAP

- MLE: maximize likelihood, 不使用 prior。
- MAP: maximize posterior, 使用 prior。

关系:

$$\text{posterior} \propto \text{likelihood} * \text{prior}$$

## 1.3 相关与回归

### Pearson correlation coefficient r

- $r = 0$ : 没有线性关系，但可能有非线性关系。
- 强度看 absolute value, 例如  $r = -0.8$  比  $r = 0.5$  更强。
- 正负号表示方向。

### Linear regression

模型:

$$\hat{y} = \beta_0 + \beta_1 x$$

- $\beta_0$ : intercept。
- $\beta_1$ : slope。

- residual = actual value - predicted value.
- residual positive: actual > predicted.

## 1.4 机器学习基本概念

### Training data

用于训练/教育 AI 系统的数据。

### Label

监督学习中的目标输出，即 desired output。

### Classification / Regression / Clustering

- Classification: 预测类别。
- Regression: 预测连续值。
- Clustering: 无监督地按相似性分组。

### Feature scaling

把不同尺度的特征变到相近尺度，常见于 kNN、SVM、k-means。

常见方法：

Min-max:  $x_{\text{scaled}} = (x - \min) / (\max - \min)$

Z-score:  $x_{\text{scaled}} = (x - \text{mean}) / \text{std}$

### Hyper-parameter

训练前人为设定，不是模型从数据中直接学出的参数。

例子：

- SVM 的 C。
- kNN 的 k。
- k-means 的 cluster number k。
- decision tree 的 max\_depth。

“两个样本之间的距离”不是 hyper-parameter。

## 1.5 PCA

Principal components:

- 是原始特征的线性组合。
- 彼此正交。
- 第一主成分解释最大方差。

基本步骤：

1. 标准化/中心化数据。
2. 求 covariance matrix。
3. 求 eigenvalues/eigenvectors。
4. 按 eigenvalue 大小选择主成分。



## 1.6 Cross Validation 与过拟合

### Cross validation

目的：

- 更稳定地估计模型泛化能力。
- 更有效利用数据。
- 用于选择 hyper-parameters。

但它不能 guarantee improved performance on independent test sets。

### Overfitting

- 训练集表现好，测试集表现差。
- 解决方式：更多数据、正则化、剪枝、降低复杂度、交叉验证选择参数。

### Underfitting

- 训练集和测试集都差。
- 模型过于简单或特征不足。

## 1.7 kNN

### 核心思想

找距离测试样本最近的  $k$  个训练样本，多数投票决定类别。

### 常见距离

Euclidean:

$$d = \sqrt{\sum (x_i - y_i)^2}$$

City-block / Manhattan:

$$d = \sum |x_i - y_i|$$

### 注意点

- $k$  太小：容易受噪声影响，过拟合。
- $k$  太大：边界过平滑，欠拟合。
- 特征尺度影响距离，所以常要 scaling。
- class imbalance 时，accuracy 可能误导。

## 1.8 Naive Bayes

### 核心假设

给定类别后，各特征条件独立：

$$P(x_1, x_2, \dots, x_d \mid C) = \text{product } P(x_i \mid C)$$

### 分类规则

选择后验概率最大的类别：

```
argmax_C P(C | x)
proportional to argmax_C P(C) * product P(x_i | C)
```

## Laplace smoothing

作用：解决 zero-probability issue。

常见形式：

```
P(feature value | class)
= (count + 1) / (class total + number of possible values)
```

## 1.9 SVM

### 核心目标

找一个 hyperplane，让两类之间的 margin 最大。

Hard-margin SVM objective:

```
minimize 1/2 ||w||^2
```

等价于 maximize margin。

### Support vectors

- 离 decision boundary 最近、决定边界位置的点。
- Lagrange multiplier  $\alpha_i > 0$  的点通常是 support vectors。
- $\alpha_i = 0$  的点模型边界没有影响。

### Soft margin

- 允许少量 margin violation。
- C 控制惩罚强度。
- C 大：更严格，少容忍错误，可能过拟合。
- C 小：更宽松，正则化更强，可能欠拟合。

### Kernel

用于处理非线性可分问题，把数据映射到更高维空间。

### Multiclass SVM

常见策略：

- OvR: one-versus-rest。
- OvO: one-versus-one，历年选择题更常出现。

## 1.10 Decision Tree 与 Ensemble

### Decision tree

树形结构，每个 internal node 测试一个 feature，每个 branch 对应测试结果，leaf node 给预测类别。

如果 leaf node 有多个类别，通常用 majority vote。

## Entropy

$$\text{Entropy}(S) = - \sum p_i \log_2(p_i)$$

越低越纯。

## Information gain

$$\text{Gain}(S, A) = \text{Entropy}(S) - \text{weighted entropy after splitting by } A$$

越大越适合作为划分属性。

## Gini impurity

$$\text{Gini}(S) = 1 - \sum p_i^2$$

越小越纯。CART 常用 Gini。

## Random forest

多个 decision trees 的集成。

## Hard voting

多个分类器投票，票数最多的类别为最终预测。

## Soft voting

综合预测概率，通常选平均/加权概率最高的类别。

## 1.11 分类评估

Confusion matrix:

TP: actual positive, predicted positive  
TN: actual negative, predicted negative  
FP: actual negative, predicted positive  
FN: actual positive, predicted negative

公式:

$$\begin{aligned}\text{Accuracy} &= (TP + TN) / (TP + TN + FP + FN) \\ \text{Precision} &= TP / (TP + FP) \\ \text{Recall} &= TP / (TP + FN) \\ F1 &= 2 * \text{Precision} * \text{Recall} / (\text{Precision} + \text{Recall})\end{aligned}$$

## AUC

- 随机分类器 expected AUC = 0.5。
- 越接近 1 越好。

## Accuracy 何时不好用

class imbalance 时不好用。例如 995 positive、5 negative，全预测 positive 也有 99.5% accuracy，但 negative recall 是 0。

## 1.12 聚类

### k-means

步骤：

1. 选 k 个初始中心。
2. 计算每个点到中心的距离。
3. 分配到最近中心。
4. 用每个 cluster 的均值更新中心。
5. 重复直到中心或分配不再变化。

优点：简单、快。

缺点：要预先指定 k；受初始中心、噪声、离群点影响；偏好球状/凸形 cluster。

### Elbow method

逐渐增大 k，观察 loss/inertia 下降曲线，选“拐点”。

### Silhouette coefficient

$$s = (b - a) / \max(a, b)$$

- a：样本到同簇其他点的平均距离。
- b：样本到最近其他簇的平均距离。
- 取值 [-1, 1]，越接近 1 越好。

### Hierarchical clustering

Agglomerative：自底向上，每个点先是一簇，每次合并最近的两个簇。

簇间距离：

- Single linkage：两簇最近点距离。
- Complete linkage / maximum distance：两簇最远点距离。
- Average linkage：所有点对平均距离。
- Centroid linkage：中心点距离。

输出 dendrogram。

## 1.13 GMM

### Gaussian Mixture Model

用多个 Gaussian components 混合表示数据分布。

每个 component 有：

- mean。
- covariance matrix。
- mixing weight。

## Diagonal covariance matrix

表示特征之间 uncorrelated，但每个特征可以有不同 variance。

## GMM vs k-means

- k-means: hard assignment, 每个点只属于一个 cluster。
- GMM: soft assignment, 每个点对每个 component 有 probability/responsibility。
- k-means 更像 spherical clusters; GMM 可以通过 covariance 描述更灵活的椭圆形分布。

## 1.14 AI 安全与类型

### Federated learning

数据留在本地设备，模型参数/更新被聚合，保护隐私。

### Data poisoning

向训练集注入恶意、错误标记的数据，降低模型性能。

### Perturbation attack

对输入做小扰动，让模型误判。

### Perception intelligence

语音识别、图像分类、情绪检测通常属于 perceptual/perception intelligence。

## 2. 计算大题模块笔记

### 模块 A: Simple Linear Regression (8 分)

#### 必背公式

模型：

$$y_{\text{hat}} = \beta_0 + \beta_1 x$$

斜率：

$$\begin{aligned}\beta_1 &= \frac{\sum((x_i - \bar{x})(y_i - \bar{y}))}{\sum((x_i - \bar{x})^2)} \\ &= S_{xy} / S_{xx}\end{aligned}$$

截距：

$$\beta_0 = \bar{y} - \beta_1 * \bar{x}$$

预测：

$$y_{\text{hat}}(\text{new}) = \beta_0 + \beta_1 * x_{\text{new}}$$

## 答题模板

1. 列出  $\bar{x}$ 、 $\bar{y}$ 。
2. 列表计算  $x_i - \bar{x}$ 、 $y_i - \bar{y}$ 、乘积、平方。
3. 求  $S_{xy}$ 、 $S_{xx}$ 。
4. 求  $\beta_1$ 、 $\beta_0$ 。
5. 写 fitted line。
6. 代入新  $x$  得预测值。

## 25-26 sample 同类题演示

数据：

$x$ : 2, 3, 4, 5, 6, 8  
 $y$ : 55, 60, 62, 68, 72, 78

计算：

```
x_bar = 28/6 = 4.667
y_bar = 395/6 = 65.833
Sxy = sum(xy) - sum(x)sum(y)/n
    = 1934 - 28*395/6
    = 90.667
Sxx = sum(x^2) - (sum(x))^2/n
    = 154 - 28^2/6
    = 23.333
beta1 = 90.667 / 23.333 = 3.886
beta0 = 65.833 - 3.886*4.667 = 47.700
```

Fitted line:

$\hat{y} = 47.700 + 3.886x$

当  $x = 7$ :

$\hat{y} = 47.700 + 3.886*7 = 74.900$

## 易错点

- residual 是 actual - predicted, 不是 predicted - actual。
- 斜率单位是  $y$  per one  $x$ 。
- 结果保留 2-3 位小数即可, 步骤要清楚。

## 模块 B: Histogram + Box Plot + Scaling (15 分)

### Histogram

题目常给一组分数，要求 equal-width bins of 10 marks。

答题步骤：

1. 写 bins，例如 10-19、20-29、30-39。
2. 统计每个 bin 的 frequency。
3. mass distribution：每个 bin 的 frequency 或 proportion。
4. cumulative distribution：从低到高累加 frequency 或 proportion。

### 25-26 sample 数据演示

数据：

12, 18, 23, 24, 27, 29, 31, 33, 35, 36,  
41, 44, 45, 48, 52, 55, 58, 62, 70, 85

按 10 分宽度分箱：

10-19: 2  
20-29: 4  
30-39: 4  
40-49: 4  
50-59: 3  
60-69: 1  
70-79: 1  
80-89: 1  
Total: 20

累计频数：

10-19: 2  
20-29: 6  
30-39: 10  
40-49: 14  
50-59: 17  
60-69: 18  
70-79: 19  
80-89: 20

考试时一定以自己数出来的 frequency 为准，最后检查 total 是否等于样本数。

### Box plot 与 outlier

五数概括：

minimum, Q1, median, Q3, maximum

IQR:

$$\text{IQR} = Q3 - Q1$$

1.5 IQR rule:

$$\text{lower fence} = Q1 - 1.5 \text{ IQR}$$

$$\text{upper fence} = Q3 + 1.5 \text{ IQR}$$

小于 lower fence 或大于 upper fence 是 outlier。

Sample 数据:

```
min = 12
Q1 = (27 + 29)/2 = 28
median = (36 + 41)/2 = 38.5
Q3 = (52 + 55)/2 = 53.5
max = 85
IQR = 53.5 - 28 = 25.5
lower fence = 28 - 1.5*25.5 = -10.25
upper fence = 53.5 + 1.5*25.5 = 91.75
```

结论:

No outliers.

## Distribution comment

答题句式:

The scores are concentrated in the lower-to-middle range, with most values below 60.  
The distribution is right-skewed because there are a few high scores such as 70 and 85.  
The box plot confirms that the upper tail is longer, but no outlier is detected by the 1.5 IQR rule.

## Scaling scheme

题目要求让 20% students achieve 70 or beyond.

20 个学生的 20% = 4 个学生。原始分数中  $\geq 70$  的只有 70、85 两人，即 10%。要让最高 4 人达到 70 分以上，至少要让第 4 高分达到 70。

排序后最高 4 个:

58, 62, 70, 85



一种简单方案:

```
new_score = min(raw_score + 12, 100)
```

这样 58 变成 70, 最高 4 人达到 70+, 且保持排名顺序。

答题句式:

To keep the ranking unchanged, I would apply a linear shift with capping:

new score = min(raw score + 12, 100).

This moves the fourth highest score, 58, to 70, so exactly the top 20% of students reach at least 70.

## 模块 C: kNN + Cross Validation (15 分)

### 必背公式

City-block / Manhattan distance:

$$d((x_1, x_2), (z_1, z_2)) = |x_1 - z_1| + |x_2 - z_2|$$

Fold accuracy:

$$\text{accuracy} = \text{correct predictions} / \text{test samples}$$

Overall CV accuracy:

$$\text{mean accuracy} = \text{average of fold accuracies}$$

### 答题模板

1. 写明每一 fold 的 training set 和 test set.
2. 对每个 test sample, 计算它到 training samples 的 city-block distance.
3. 找 nearest neighbour.
4. 写 predicted label.
5. 判断 correct/incorrect.
6. 每 fold 算 accuracy.
7. 取 fold accuracies 的平均.

### 25-26 sample 同类题演示

使用 1-NN, fixed folds:

Fold 1 test: {1, 2, 3, 4}

Fold 2 test: {5, 6, 7, 8}

Fold 3 test: {9, 10, 11, 12}

可整理成表格作答：

Fold 1:

Sample 1 -> nearest 10 or 12, distance 2, predicted 0, correct

Sample 2 -> nearest 12, distance 1, predicted 0, correct

Sample 3 -> nearest 10 or 12, distance 1, predicted 0, correct

Sample 4 -> nearest 12, distance 1, predicted 0, correct

Accuracy =  $4/4 = 1.00$

Fold 2:

Sample 5 -> nearest 9, distance 2, predicted 1, correct

Sample 6 -> nearest 9, distance 3, predicted 1, correct

Sample 7 -> nearest 9, distance 3, predicted 1, correct

Sample 8 -> nearest 9, distance 5, predicted 1, correct

Accuracy =  $4/4 = 1.00$

Fold 3:

Sample 9 -> nearest 5, distance 2, predicted 1, correct

Sample 10 -> nearest 3, distance 1, predicted 0, correct

Sample 11 -> nearest 4, distance 2, predicted 0, incorrect

Sample 12 -> nearest 3 or 4, distance 1, predicted 0, correct

Accuracy =  $3/4 = 0.75$

Overall CV accuracy =  $(1.00 + 1.00 + 0.75) / 3 = 0.9167 = 91.67\%$

## 易错点

- test fold 里的样本不能作为自己的 training neighbour。
- 题目若规定 tie-breaking, 必须照做。
- 1-NN 不投票; k=3、5 才多数投票。
- City-block 不要开根号。

## 模块 D: Naive Bayes 手算 (旧题高频)

### 答题模板

若要预测 class Yes/No:

$\text{score(Yes)} = P(\text{Yes}) * P(\text{feature1} \mid \text{Yes}) * P(\text{feature2} \mid \text{Yes}) * \dots$

$\text{score(No)} = P(\text{No}) * P(\text{feature1} \mid \text{No}) * P(\text{feature2} \mid \text{No}) * \dots$

比较两个 score, 较大者为预测类别。

### 连续条件如何处理

历年 golf table 里会出现:

- humidity lower than 65%

- humidity higher than 65%
- wind speed higher than 2.0 m/s

做法：先把条件转为计数事件，例如在 Yes 类里统计 humidity < 65 的样本数。

## 答题句式

Since score(Yes) > score(No), the posterior probability of Yes is larger.  
Therefore, the predicted preference is Yes.

## Laplace smoothing

若某个条件频数为 0，要使用 Laplace correction:

$(\text{count} + 1) / (\text{class total} + \text{number of possible values})$

## 模块 E: k-means 手算

### 答题模板

1. 写初始 centroids。
2. 对每个点计算到各中心距离。
3. 分配到最近中心。
4. 对每个 cluster 取均值，更新 centroid。
5. 若题目只要求 one iteration，到这里结束。

### 23-24 sample 同类题演示

点：

A=(2,3), B=(3,4), C=(10,15), D=(15,12), E=(10,10), F=(10,14)  
Z1=(2,3), Z2=(10,15)

City-block distance 后分配：

Cluster 1: A, B  
Cluster 2: C, D, E, F

新中心：

$Z1' = ((2+3)/2, (3+4)/2) = (2.5, 3.5)$   
 $Z2' = ((10+15+10+10)/4, (15+12+10+14)/4)$   
 $= (11.25, 12.75)$

## 模块 F: Hierarchical Clustering / Dendrogram

### 答题模板

1. 计算所有点对距离。
2. 每次合并距离最小的两个 cluster。
3. 如果是 maximum distance / complete linkage, 两个 cluster 的距离取所有跨簇点对中的最大距离。
4. 记录每次合并高度。
5. 画 dendrogram。

### 23-24 final 同类题提示

点:

$A=(1,2)$ ,  $B=(2,0)$ ,  $C=(1,0)$ ,  $D=(1,1)$ ,  $E=(4,0)$

City-block 点对距离:

$A-B=3$ ,  $A-C=2$ ,  $A-D=1$ ,  $A-E=5$   
 $B-C=1$ ,  $B-D=2$ ,  $B-E=2$   
 $C-D=1$ ,  $C-E=3$   
 $D-E=4$

若按 complete linkage, 可出现 tie。一个合理合并顺序是:

1. merge A and D at distance 1
2. merge B and C at distance 1
3. merge (B,C) and E at distance 3
4. merge (A,D) and (B,C,E) at distance 5

考试遇到 tie 时, 要写清楚 chosen tie-breaking; 不同 tie-breaking 可能 dendrogram 形状略不同, 但距离计算逻辑必须一致。

## 3. 答题模块: GMM Model Selection vs Cross Validation (6 分)

### 必背答案

Model selection criteria such as AIC and BIC evaluate a GMM by combining goodness of fit and model complexity. They use the likelihood of the fitted model and add a penalty for the number of parameters. AIC usually focuses more on predictive performance, while BIC has a stronger penalty for complex models and often prefers simpler models when the sample size is large.

Cross validation evaluates a GMM by splitting data into training and validation sets. The model is fitted on training data and evaluated on held-out data, usually by validation log-likelihood. It directly estimates generalisation performance, but it is more computationally expensive and depends on the split.

Assumptions: AIC/BIC assume maximum-likelihood fitting, independent samples, large-sample approximation, and regular model conditions. BIC also assumes that the true model is among the candidate models. Cross validation assumes that training and validation data are independently drawn from the same distribution, so validation performance represents future unseen data.

## 中文理解版

- AIC/BIC: 看“拟合得好不好”加“模型复杂不复杂”。
- AIC penalty 较轻, 更偏预测。
- BIC penalty 较重, 更偏简单模型。
- Cross validation: 直接用留出的数据测试泛化能力。
- AIC/BIC 更快, CV 更直观但更费计算。

## 6 分答题结构

1. Define AIC/BIC: likelihood + complexity penalty.
2. Define CV: train/validation split, held-out log-likelihood.
3. Compare: AIC/BIC faster and assumption-based; CV computational but direct.
4. Assumptions: iid data; AIC/BIC large-sample/MLE; BIC true model among candidates; CV same distribution between train and validation.

## 4. 大题真题/样题与完整版英文解答

使用方式: 先遮住答案自己手算, 再对照 English solution。考试只接受英文答案, 所以英文部分尽量按可直接写进答题纸的风格整理; 中文解释放在后面帮助理解。

### 4.1 25-26 Sample Q29: Simple Linear Regression (8 marks)

#### Question

A tutor wants to model the relationship between weekly study time  $x$  and exam score  $y$  using simple linear regression.

$x$ : 2, 3, 4, 5, 6, 8  
 $y$ : 55, 60, 62, 68, 72, 78

Compute the least-squares estimates of the regression coefficients, write the fitted regression line, and predict  $y$  when  $x = 7$ .

#### Full English Answer

First, calculate the sample means:

$$\begin{aligned} \bar{x} &= (2 + 3 + 4 + 5 + 6 + 8) / 6 = 28 / 6 = 4.667 \\ \bar{y} &= (55 + 60 + 62 + 68 + 72 + 78) / 6 = 395 / 6 = 65.833 \end{aligned}$$

For simple linear regression, the least-squares slope is:

$$\text{beta1\_hat} = S_{xy} / S_{xx}$$

where:

$$\begin{aligned} S_{xy} &= \sum(x_i y_i) - \frac{\sum(x_i)\sum(y_i)}{n} \\ S_{xx} &= \sum(x_i^2) - \frac{(\sum(x_i))^2}{n} \end{aligned}$$

Now:

$$\begin{aligned} \sum(x_i y_i) &= 2*55 + 3*60 + 4*62 + 5*68 + 6*72 + 8*78 \\ &= 1934 \end{aligned}$$

$$\begin{aligned} \sum(x_i^2) &= 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 8^2 \\ &= 154 \end{aligned}$$

$$\begin{aligned} S_{xy} &= 1934 - (28*395)/6 \\ &= 1934 - 1843.333 \\ &= 90.667 \end{aligned}$$

$$\begin{aligned} S_{xx} &= 154 - 28^2/6 \\ &= 154 - 130.667 \\ &= 23.333 \end{aligned}$$

Therefore:

$$\begin{aligned} \text{beta1\_hat} &= 90.667 / 23.333 = 3.886 \\ \text{beta0\_hat} &= \bar{y} - \text{beta1\_hat} * \bar{x} \\ &= 65.833 - 3.886 * 4.667 \\ &= 47.700 \end{aligned}$$

The fitted regression line is:

$$\hat{y} = 47.700 + 3.886x$$

When  $x = 7$ :

$$\hat{y} = 47.700 + 3.886 * 7 = 74.902$$

So the predicted exam score is approximately:

$$74.9$$

## 中文解释

这题公式考试会给，但你要会代入。先算  $\bar{x}$  和  $\bar{y}$  的均值，再算  $S_{xy}$  和  $S_{xx}$ 。斜率表示每多学习 1 小时，预测分数大约增加 3.886 分。截距是  $x=0$  时模型预测值。最后代入  $x=7$  得到约 74.9 分。

## 4.2 25-26 Sample Q31: Histogram, Box Plot, Outliers, Scaling (15 marks)

### Question

The exam scores of 20 students are:

12, 18, 23, 24, 27, 29, 31, 33, 35, 36,  
41, 44, 45, 48, 52, 55, 58, 62, 70, 85

Construct a histogram using equal-width bins of 10 marks, draw a box plot using the five-number summary, identify outliers using the 1.5 IQR rule, and propose a scaling scheme that makes 20% of students achieve 70 marks or beyond.

### Full English Answer

Using equal-width bins of 10 marks, the mass distribution is:

| Bin   | Frequency | Relative frequency |
|-------|-----------|--------------------|
| 10-19 | 2         | $2/20 = 0.10$      |
| 20-29 | 4         | $4/20 = 0.20$      |
| 30-39 | 4         | $4/20 = 0.20$      |
| 40-49 | 4         | $4/20 = 0.20$      |
| 50-59 | 3         | $3/20 = 0.15$      |
| 60-69 | 1         | $1/20 = 0.05$      |
| 70-79 | 1         | $1/20 = 0.05$      |
| 80-89 | 1         | $1/20 = 0.05$      |

The cumulative distribution is:

| Bin   | Cumulative frequency | Cumulative relative frequency |
|-------|----------------------|-------------------------------|
| 10-19 | 2                    | 0.10                          |
| 20-29 | 6                    | 0.30                          |
| 30-39 | 10                   | 0.50                          |
| 40-49 | 14                   | 0.70                          |
| 50-59 | 17                   | 0.85                          |
| 60-69 | 18                   | 0.90                          |
| 70-79 | 19                   | 0.95                          |
| 80-89 | 20                   | 1.00                          |

For the box plot, the data are already sorted. Since there are 20 observations:

Minimum = 12  
Median =  $(36 + 41) / 2 = 38.5$   
 $Q1 = (27 + 29) / 2 = 28$   
 $Q3 = (52 + 55) / 2 = 53.5$   
Maximum = 85

Therefore:

$$\text{IQR} = Q3 - Q1 = 53.5 - 28 = 25.5$$

$$\text{Lower fence} = Q1 - 1.5 \cdot \text{IQR} = 28 - 1.5 \cdot 25.5 = -10.25$$

$$\text{Upper fence} = Q3 + 1.5 \cdot \text{IQR} = 53.5 + 1.5 \cdot 25.5 = 91.75$$

All scores are between `-10.25` and `91.75`, so there are no outliers according to the 1.5 IQR rule.

The distribution is concentrated in the lower-to-middle score range, and most scores are below 60. The higher scores such as 70 and 85 make the distribution slightly right-skewed, but 85 is not an outlier by the 1.5 IQR rule.

There are 20 students, so 20% of the students means:

$$20 \cdot 20\% = 4 \text{ students}$$

To make 4 students achieve 70 or above while keeping the ranking unchanged, the fourth highest score should be increased to 70. The four highest original scores are:

58, 62, 70, 85

Thus, one reasonable scaling scheme is:

$$\text{new score} = \min(\text{original score} + 12, 100)$$

Under this scheme, the score 58 becomes 70, so exactly the top 20% of students achieve at least 70 marks. The `min` function also ensures that no score exceeds 100.

### 中文解释

这题最容易错在分箱计数和累计频数。每个 bin 的频数加起来必须等于 20。箱线图按五数概括画：最小值、Q1、中位数、Q3、最大值。异常值用上下 fence 判断。最后 scaling 不需要唯一答案，只要说清楚方案合理即可；这里加 12 分让第 4 高分 58 变成 70，同时用 `min(...,100)` 防止超过 100。

## 4.3 25-26 Sample Q32: 3-fold Cross-validation for 1-NN (15 marks)

### Question

Use 3-fold cross-validation to evaluate a 1-NN classifier with city-block distance:

$$d(x_i, x_j) = |x_{i1} - x_{j1}| + |x_{i2} - x_{j2}|$$

The fixed test folds are:

Fold 1: {1, 2, 3, 4}

Fold 2: {5, 6, 7, 8}

Fold 3: {9, 10, 11, 12}

Dataset:



| ID | x1 | x2 | y |
|----|----|----|---|
| 1  | 2  | 1  | 0 |
| 2  | 3  | 1  | 0 |
| 3  | 2  | 2  | 0 |
| 4  | 4  | 2  | 0 |
| 5  | 6  | 5  | 1 |
| 6  | 7  | 5  | 1 |
| 7  | 6  | 6  | 1 |
| 8  | 8  | 6  | 1 |
| 9  | 5  | 4  | 1 |
| 10 | 1  | 2  | 0 |
| 11 | 5  | 3  | 1 |
| 12 | 3  | 2  | 0 |

### Full English Answer

For each fold, the test samples are classified using the remaining 8 samples as the training set.

For Fold 1, the test set is {1, 2, 3, 4} and the training set is {5, 6, 7, 8, 9, 10, 11, 12}.

Sample 1: nearest neighbour = 10 or 12, distance = 2, predicted  $y = 0$ , correct  
Sample 2: nearest neighbour = 12, distance = 1, predicted  $y = 0$ , correct  
Sample 3: nearest neighbour = 10 or 12, distance = 1, predicted  $y = 0$ , correct  
Sample 4: nearest neighbour = 12, distance = 1, predicted  $y = 0$ , correct

Therefore:

Fold 1 accuracy =  $4/4 = 1.00$

For Fold 2, the test set is {5, 6, 7, 8} and the training set is {1, 2, 3, 4, 9, 10, 11, 12}.

Sample 5: nearest neighbour = 9, distance = 2, predicted  $y = 1$ , correct  
Sample 6: nearest neighbour = 9, distance = 3, predicted  $y = 1$ , correct  
Sample 7: nearest neighbour = 9, distance = 3, predicted  $y = 1$ , correct  
Sample 8: nearest neighbour = 9, distance = 5, predicted  $y = 1$ , correct

Therefore:

Fold 2 accuracy =  $4/4 = 1.00$

For Fold 3, the test set is {9, 10, 11, 12} and the training set is {1, 2, 3, 4, 5, 6, 7, 8}.

Sample 9: nearest neighbour = 5, distance = 2, predicted  $y = 1$ , correct  
Sample 10: nearest neighbour = 3, distance = 1, predicted  $y = 0$ , correct  
Sample 11: nearest neighbour = 4, distance = 2, predicted  $y = 0$ , incorrect  
Sample 12: nearest neighbour = 2, 3, or 4, distance = 1, predicted  $y = 0$ , correct

Therefore:

$$\text{Fold 3 accuracy} = 3/4 = 0.75$$

The overall cross-validation accuracy is the mean of the three fold accuracies:

$$\begin{aligned}\text{Overall CV accuracy} &= (1.00 + 1.00 + 0.75) / 3 \\ &= 0.9167 \\ &= 91.67\%\end{aligned}$$

Therefore, the 3-fold cross-validation accuracy of the 1-NN classifier is approximately 91.67%.

### 中文解释

这题关键是：每一折的测试集不能参与训练。1-NN 只找一个最近邻，不需要多数投票。距离是 city-block，所以只加绝对值差，不开根号。Sample 11 被错分，因为它最近的训练样本是 4，标签为 0，但真实标签是 1。

## 4.4 25-26 Sample Q30: GMM Model Selection Criteria vs Cross Validation (7 marks)

### Question

Compare model selection criteria and cross validation when used to evaluate a Gaussian Mixture Model. What assumptions are used for model selection criteria and cross validation?

### Full English Answer

Model selection criteria, such as AIC and BIC, evaluate a Gaussian Mixture Model by combining goodness of fit and model complexity. They use the likelihood of the fitted model and add a penalty for the number of parameters. AIC is usually more focused on predictive performance, while BIC gives a stronger penalty to complex models, especially when the sample size is large. Therefore, BIC often prefers simpler models than AIC.

Cross validation evaluates a GMM by splitting the dataset into training and validation subsets. The GMM is fitted on the training data, and then its performance is measured on the validation data, usually using validation log-likelihood. This gives a more direct estimate of generalisation performance on unseen data, but it is more computationally expensive because the model must be trained multiple times.

The assumptions are different. AIC and BIC assume that the model is fitted by maximum likelihood, the samples are independent and identically distributed, and the sample size is sufficiently large for the asymptotic approximation to be meaningful. BIC also assumes that the true model is among the candidate models. Cross validation assumes that the training and validation data are drawn independently from the same underlying distribution, so that validation performance is representative of future unseen data.

In summary, AIC and BIC are faster and formula-based, but rely more strongly on modelling assumptions. Cross validation is more computationally expensive, but it directly tests the model on held-out data.

### 中文解释

AIC/BIC 是 “拟合好不好 + 模型复杂度惩罚” 。AIC 惩罚较轻, BIC 惩罚较重, 所以 BIC 更偏向简单模型。Cross validation 是把数据切开, 用训练集拟合、验证集评估, 更接近真实泛化表现, 但计算更贵。回答题一定要写 assumptions: iid、maximum likelihood、大样本近似、训练/验证来自同一分布。

## 4.5 22-23 Final: Naive Bayes Golf Table (14 marks)

### Question

Using Naive Bayes, predict the golf player's preference under the condition:

Outlook = Rainy  
Humidity < 65%

The table has 10 samples. There are 5 Yes samples and 5 No samples.

### Full English Answer

We compare the Naive Bayes scores for the two possible classes, Yes and No .

For class Yes :

$P(\text{Yes}) = 5/10$   
 $P(\text{Rainy} \mid \text{Yes}) = 3/5$   
 $P(\text{Humidity} < 65\% \mid \text{Yes}) = 4/5$

Therefore:

$\text{Score}(\text{Yes}) = P(\text{Yes}) * P(\text{Rainy} \mid \text{Yes}) * P(\text{Humidity} < 65\% \mid \text{Yes})$   
 $= (5/10) * (3/5) * (4/5)$   
 $= 0.24$

For class No :

$P(\text{No}) = 5/10$   
 $P(\text{Rainy} \mid \text{No}) = 2/5$   
 $P(\text{Humidity} < 65\% \mid \text{No}) = 1/5$

Therefore:

$\text{Score}(\text{No}) = P(\text{No}) * P(\text{Rainy} \mid \text{No}) * P(\text{Humidity} < 65\% \mid \text{No})$   
 $= (5/10) * (2/5) * (1/5)$   
 $= 0.04$

Since:

$\text{Score}(\text{Yes}) > \text{Score}(\text{No})$

the predicted preference is:

Yes

### 中文解释

Naive Bayes 大题不用算真正归一化后的后验概率，因为分母  $P(x)$  对两个类别相同，比较分子即可。注意每个条件概率都要在对应类别内部计数：比如  $P(\text{Rainy} \mid \text{Yes})$  是 Yes 样本里 Rainy 的比例。

## 4.6 22-23 Resit: Naive Bayes Golf Table (14 marks)

### Question

Using Naive Bayes, predict the golf player's preference under the condition:

Outlook = Sunny  
Wind speed > 2.0 m/s

### Full English Answer

Again, compare the Naive Bayes scores for Yes and No .

For class Yes :

$P(\text{Yes}) = 5/10$   
 $P(\text{Sunny} \mid \text{Yes}) = 2/5$   
 $P(\text{Wind speed} > 2.0 \mid \text{Yes}) = 1/5$

Thus:

$\text{Score}(\text{Yes}) = (5/10) * (2/5) * (1/5)$   
 $= 0.04$

For class No :

$P(\text{No}) = 5/10$   
 $P(\text{Sunny} \mid \text{No}) = 3/5$   
 $P(\text{Wind speed} > 2.0 \mid \text{No}) = 3/5$

Thus:

$\text{Score}(\text{No}) = (5/10) * (3/5) * (3/5)$   
 $= 0.18$

Since:

$\text{Score}(\text{No}) > \text{Score}(\text{Yes})$

the predicted preference is:

No

### 中文解释

这道题和上一题完全同套路，只是条件换成了 Sunny 和 Wind speed > 2.0。最后比较两个 score，No 更大，所以预测 No。

## 4.7 23-24 Sample: One Iteration of k-means (14 marks)

### Question

Perform one iteration of k-means with  $k = 2$ , using city-block distance. Initial centroids:

$z_1 = (2, 3)$

$z_2 = (10, 15)$

Points:

$A = (2, 3)$ ,  $B = (3, 4)$ ,  $C = (10, 15)$ ,

$D = (15, 12)$ ,  $E = (10, 10)$ ,  $F = (10, 14)$

### Full English Answer

Using city-block distance:

$$d((x_1, x_2), (z_1, z_2)) = |x_1 - z_1| + |x_2 - z_2|$$

Calculate the distance from each point to the two centroids:

Point A:

$$d(A, Z1) = |2-2| + |3-3| = 0$$

$$d(A, Z2) = |2-10| + |3-15| = 20$$

=> A belongs to Cluster 1

Point B:

$$d(B, Z1) = |3-2| + |4-3| = 2$$

$$d(B, Z2) = |3-10| + |4-15| = 18$$

=> B belongs to Cluster 1

Point C:

$$d(C, Z1) = |10-2| + |15-3| = 20$$

$$d(C, Z2) = |10-10| + |15-15| = 0$$

=> C belongs to Cluster 2

Point D:

$$d(D, Z1) = |15-2| + |12-3| = 22$$

$$d(D, Z2) = |15-10| + |12-15| = 8$$

=> D belongs to Cluster 2

Point E:

$$d(E, Z1) = |10-2| + |10-3| = 15$$

$$d(E, Z2) = |10-10| + |10-15| = 5$$

=> E belongs to Cluster 2

Point F:

$$d(F, Z1) = |10-2| + |14-3| = 19$$

$$d(F, Z2) = |10-10| + |14-15| = 1$$

=> F belongs to Cluster 2

Thus:

Cluster 1 = {A, B}

Cluster 2 = {C, D, E, F}

Update the centroids by taking the mean of the points in each cluster:

$$\begin{aligned} Z1_{\text{new}} &= ((2+3)/2, (3+4)/2) \\ &= (2.5, 3.5) \end{aligned}$$

$$\begin{aligned} Z2_{\text{new}} &= ((10+15+10+10)/4, (15+12+10+14)/4) \\ &= (45/4, 51/4) \\ &= (11.25, 12.75) \end{aligned}$$

After one iteration, the new centroids are:

$Z1 = (2.5, 3.5)$   
 $Z2 = (11.25, 12.75)$

### 中文解释

k-means 手算题有两步：先分配，再更新中心。题目说 one iteration，就做到新中心为止，不需要继续迭代到收敛。距离用 city-block，所以绝对值相加。

## 4.8 23-24 Final: Agglomerative Clustering with Complete Linkage (14 marks)

### Question

Given:

$A = (1, 2)$ ,  $B = (2, 0)$ ,  $C = (1, 0)$ ,  $D = (1, 1)$ ,  $E = (4, 0)$

Cluster the data points by agglomerative clustering with maximum city-block distance and draw the cluster dendrogram.

### Full English Answer

First, compute the city-block distances between all pairs:

$d(A,B) = |1-2| + |2-0| = 3$   
 $d(A,C) = |1-1| + |2-0| = 2$   
 $d(A,D) = |1-1| + |2-1| = 1$   
 $d(A,E) = |1-4| + |2-0| = 5$   
  
 $d(B,C) = |2-1| + |0-0| = 1$   
 $d(B,D) = |2-1| + |0-1| = 2$   
 $d(B,E) = |2-4| + |0-0| = 2$   
  
 $d(C,D) = |1-1| + |0-1| = 1$   
 $d(C,E) = |1-4| + |0-0| = 3$   
  
 $d(D,E) = |1-4| + |1-0| = 4$

Because the question uses maximum city-block distance, the distance between two clusters is defined by complete linkage:

$D(\text{cluster 1, cluster 2}) = \text{maximum pairwise distance between points in the two clusters}$

The smallest pairwise distance is 1. There is a tie. One valid merging order is:

Step 1: Merge A and D at distance 1.  
Step 2: Merge B and C at distance 1.

Now the clusters are:

{A,D}, {B,C}, {E}

Compute complete-linkage distances:

$$\begin{aligned}D(\{A,D\}, \{B,C\}) &= \max(d(A,B), d(A,C), d(D,B), d(D,C)) \\&= \max(3, 2, 2, 1) \\&= 3\end{aligned}$$

$$\begin{aligned}D(\{A,D\}, \{E\}) &= \max(d(A,E), d(D,E)) \\&= \max(5, 4) \\&= 5\end{aligned}$$

$$\begin{aligned}D(\{B,C\}, \{E\}) &= \max(d(B,E), d(C,E)) \\&= \max(2, 3) \\&= 3\end{aligned}$$

The next minimum complete-linkage distance is 3. Merge {B,C} and {E} :

Step 3: Merge {B,C} and {E} at distance 3.

Now the clusters are:

{A,D}, {B,C,E}

The final complete-linkage distance is:

$$\begin{aligned}D(\{A,D\}, \{B,C,E\}) &= \max(d(A,B), d(A,C), d(A,E), d(D,B), d(D,C), d(D,E)) \\&= \max(3, 2, 5, 2, 1, 4) \\&= 5\end{aligned}$$

Therefore:

Step 4: Merge {A,D} and {B,C,E} at distance 5.

The dendrogram should show:

A and D merge at height 1.  
B and C merge at height 1.  
{B,C} merges with E at height 3.  
{A,D} merges with {B,C,E} at height 5.

Due to ties at distance 1 and 3, another dendrogram with a different tie-breaking order may also be acceptable if the complete-linkage distances are calculated correctly.

**中文解释**



题目里的 “maximum city-block distance” 就是 complete linkage: 两个簇之间的距离取跨簇所有点对距离的最大值。层次聚类题一定要写每次合并的高度, 因为 dendrogram 的纵轴就是这个高度。遇到 tie 不怕, 但要说明你采用的合并顺序。

## 4.9 22-23 Final: kNN Golf Table with City-block Distance (10 marks)

### Question

Using kNN with  $k = 3$ , predict the preference under:

Outlook = Sunny  
Humidity = 65%  
Wind speed = 3.0 m/s

Use city-block distance on humidity and wind speed:

$$d = |\text{humidity difference}| + |\text{wind speed difference}|$$

### Full English Answer

The query point is:

$$Q = (65, 3.0)$$

Calculate city-block distance from the query to the training samples:

Rainy, 80, 0.5, Yes:  $d = |65-80| + |3.0-0.5| = 17.5$   
Rainy, 40, 0.2, Yes:  $d = |65-40| + |3.0-0.2| = 27.8$   
Rainy, 50, 5.0, No:  $d = |65-50| + |3.0-5.0| = 17.0$   
Rainy, 50, 0.2, Yes:  $d = |65-50| + |3.0-0.2| = 17.8$   
Rainy, 75, 4.0, No:  $d = |65-75| + |3.0-4.0| = 11.0$   
Sunny, 70, 5.0, No:  $d = |65-70| + |3.0-5.0| = 7.0$   
Sunny, 75, 0.4, No:  $d = |65-75| + |3.0-0.4| = 12.6$   
Sunny, 80, 0.1, No:  $d = |65-80| + |3.0-0.1| = 17.9$   
Sunny, 50, 0.2, Yes:  $d = |65-50| + |3.0-0.2| = 17.8$   
Sunny, 40, 4.0, Yes:  $d = |65-40| + |3.0-4.0| = 26.0$

The three nearest neighbours are:

Sunny, 70, 5.0, No with distance 7.0  
Rainy, 75, 4.0, No with distance 11.0  
Sunny, 75, 0.4, No with distance 12.6

All three nearest neighbours have label **No**. Therefore, by majority vote:

Predicted preference = No

### 中文解释

这题虽然题干提到 Sunny day, 但给定距离公式只用 humidity 和 wind speed, 所以通常按这两个数值特征算距离。若老师要求先筛选 Sunny 样本, 预测仍然是 No。答题时最重要的是把距离表列清楚, 然后写最近的 3 个邻居和多数投票。

## 5. 选择题速记清单

看到这些词, 优先选这些概念

```
statistic -> sample
parameter -> population
voluntary response -> sampling bias
normal distribution -> continuous and symmetric
CLT -> population need not be normal
confidence interval -> range of values estimating parameter
MLE -> no prior
MAP -> uses prior
 $r=0$  -> no linear relationship
largest  $|r|$  -> strongest linear relationship
positive residual -> actual > predicted
intercept ->  $\beta_0$ 
kNN common distance -> Euclidean
k-means steps -> assign samples and update centroids
Laplace smoothing -> fix zero probability
SVM  $\alpha=0$  -> no effect on model
hard-margin SVM objective -> maximize margin
 $p\text{-value} > \alpha$  -> fail to reject  $H_0$ 
decision tree mixed leaf -> majority vote
F1 -> harmonic mean of precision and recall
random classifier AUC -> 0.5
overfitting solution -> more data / regularization / pruning
privacy across devices -> federated learning
malicious mislabeled training data -> data poisoning
speech/image/emotion recognition -> perception intelligence
Naive Bayes naive assumption -> conditional independence
diagonal covariance in GMM -> uncorrelated features with variable variances
```

## 6. 考前备考方案

### 第 1 轮: 1 天, 搭框架

目标: 选择题至少稳定拿 40+/56。

任务:

1. 背第 1 章选择题高频知识点。
2. 把第 4 章速记清单抄一遍。
3. 做 25-26 sample 的 28 个选择题。

4. 错题按模块标记：统计、NB、SVM、评估、聚类、AI安全。

## 第2轮：1-2天，专攻计算题

目标：3道计算题至少拿30+/38。

必须手写练习：

1. Linear regression：完整做一遍 sample Q29。
2. Histogram/boxplot：完整做一遍 sample Q31。
3. kNN CV：完整做一遍 sample Q32。
4. k-means：做 23-24 sample 的 one iteration。
5. Naive Bayes：做 22-23 final/resit golf table。
6. Hierarchical clustering：做 23-24 final dendrogram。

## 第3轮：半天，准备答题

目标：6分答题拿4-6分。

背三段英文：

1. GMM AIC/BIC vs cross validation。
2. Accuracy limitation under class imbalance。
3. Overfitting definition and solutions。

## 第4轮：考前2小时

只看：

1. 第4章选择题速记清单。
2. 线性回归公式。
3. box plot fences。
4. kNN CV 答题模板。
5. GMM 答题模板。

## 7. 考试答题提醒

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1. 计算题必须写步骤，不能只写答案。
2. 距离题先确认是 Euclidean 还是 City-block。
3. Cross validation 中 test fold 不能参与训练。
4. Box plot 一定检查 total frequency 是否等于样本数。
5. 答题尽量用英文关键词：likelihood、complexity penalty、held-out data、iid、generalisation。
6. Python 相关题不用投入，但选择题里如果出现 virtual environment / API 之类，按常识做即可，不作为主复习内容。